NANO SPEEDLOADER

USER GUIDE



Summary

1. Introduction 3

2. Requirements 4

3. Operating Guide 5

4. Assembly/Disassembly Guide 9

5. Software Update 12

Introduction



Thanks for buying our product, This guide will show you the menu options and how to update the NANO SpeedLoader,

In many ways this object will become your all time companion on the Airsoft field, while on the safe zone but also on the field due to his pocket side, you can carry it with you all day and reload your mags in few seconds.

The key advantages of this product is the integration of an ESP32 Ship, providing more customization options, language support, animations, and a motor controller board that allows users to adjust the speed for enhanced control and precision.

This project is developed with an evolving Arduino code, continuously enhanced based on user feedback. As an enthusiast this solution may have limitations from various perspectives. All enhancement feedbacks are welcome to refine and optimize the design.

Requirements & Prerequisites

Necessary Tools:

- Standard screwdriver set
- Hex keys (3 & 2mm)

1 Operating Guide

Customized on-screen animations. The menu is designed to be user-friendly by displaying icons of the most iconic replicas we see on Airsoft fields. For each preset the user can choose number of bbs to be loaded, the speed, the name and even the icon. Users can interact with the Speedloader using the four arrows and the trigger for intuitive navigation and selection.

Up and Down :

To navigate in menu, settings and preferences. Long press will move the selection faster.



Back / Power button :

Long press to turn **ON** or **OFF** the Speedloader.

Single press : back to previous menus during menu navigation.

On main screen it has two functions: **First press** to **reset the current counter**, second and following press will **increase brightness** of the screen 💡

On the rename menu, it delete the last character, multiple press will erase the full preset name.

Right / Ok Button :

This button let you open any preset in edit mode or confirm a choice. Some menus like global counter and Rename require a long press to save choice.

In rename screen : Single press will move the selection to the next character. Long press will save the preset name.

Embedded Features:



Startup Sequence: The firmware starts on welcome screen displayed for 2 seconds, you will hear a buzz sound from the gearbox that will shake few times to avoid bb jam because of long storage.

Anti-Jam mode:

ℹ️ Depending on print material, temperature change, humidity, storage duration, Gearbox mechanism might Jam. Startup sequence can be extended to solve that.

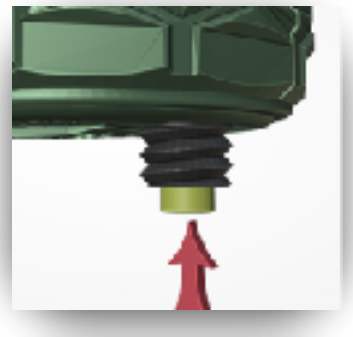


➡️ hold down arrow⬇️ Pressed when starting up the Nano Speedloader. The servo will shake back an forth in order to extract all the jammed BB inside the mechanism. The procedure until the button is released.

BB Weight display :

Help to track last bb weight used, it is possible to edit the weight only during the startup using the two **UP** and **DOWN** buttons when the weight screen is displayed.

Combat start :



Long Press on the trigger will **power ON** the Speedloader directly in loading mode , starting on the last preset used. And thus to make it easy to reload with a single action.

When you release the trigger the loader goes back to main screen, you can resume the the feeding any time, and continue on the same counter.



ℹ️ The feeding counter and timer will reset after **1 min** if not used. If you would like to reset the counter sooner, you can press on back butto⬅️

HOME Screen :



Main screen of the speedloader, allow to display up to 19 customizable presets and access the settings submenu. Each preset has its own submenu that allows fine-tuning of preset parameters.

Partial counter and **BB weight** are displayed on top of the screen, this counter can be reset manually by the user.

Loading counter : by default the lower left corner shows the function of the left button, when a reloading is initiated, it displays the loading counter, you will be able to see how many bbs got loaded in the mag. If not used, counter and timer reset after 1 min.

Battery : display the charge level, this reading relied on the battery voltage, for smooth and accurate ready make sure to use only 7.4v 2S battery.

Backlight Control: Adjust the screen brightness to match your lighting conditions. The system offers 10 levels of brightness, ranging from almost off to fully exposed, allowing optimal visibility in different environments.



ℹ Last preset used for more than 1 min will be saves as a default startup preset, tu allow Trigger start mode to start on the last preset used.

ℹ Each mode can be edited to match user needs (except the free-mode)

- The number of BBs to load
- The loading speed: to adjust the feeding speed
- The display name of the preset
- Change the icon displayed among the 18 icons of the library

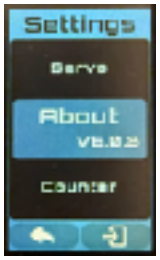
Loading Screen :



The loading screen is displayed only when the trigger is pressed, it shows few insights about the loading process

- On going loaded bbs and the loading rate.
- Loading duration

Settings Menu



Energy Saving: Among the options provided, there is a sub-menu that allows users to select the time before the board automatically shuts down to save battery



Languages: The Software is provided in 6 languages, translations have been made on google translate, don't hesitate if you have suggestions.

Servo Calibration: This setting menue allow you to fine tune the minimum and maximum power send to the servo, and thus to have a better power management.



Counter : This sub-menu keeps track of loader usage, it shows two counters

- Global counter : the total bb loaded, this counter cannot be reset.
- Partial counter : this counter is the same displayed on the home screen, User can reset this counter holding right button pressed for 3 seconds.



Combat mode: This sub-menu allow user to choose if the trigger can start the loader or if the device should ignore this action. Option enabled by default.

Preset : This sub-menu allow user to choose how many presets to display on the home menu, choice goes from 3 to 20, at least user can Have Settings, Free mode and a custom preset, maximum is 19 presets + Settings menu.



About : This last menu gives an information about the current version of the software used. It is also used to start OTA (over the air) updates. Long press on right arrow to restart in Update mode. Please note that the OTA mode is available from the V6 of the firmware, updating former version, an update via USB is required. Please refer to the Update section to have more details.

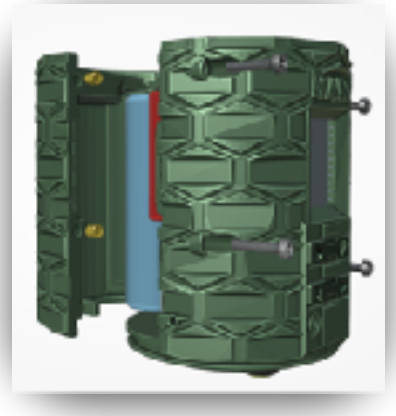
2 | **Assembly/ Disassembly** | **15 Mins**

Disassembly

Opening the loader in 5 steps:

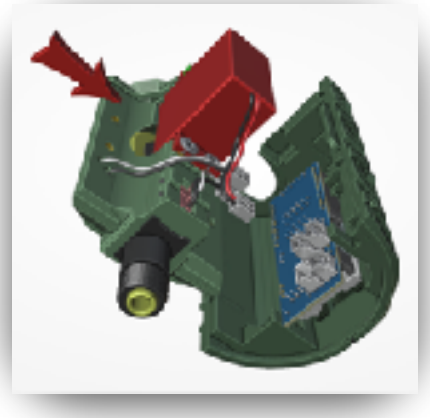


➡️ Remove the side screw to release the tank, turn the tank in CCW rotation to unlock.

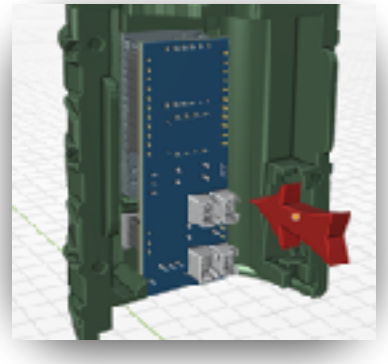


➡️ Remove the 4x M3 front hex screws to allow the loader to open.

➡️ Use a flat handle to put levers on the side edges, you can open the loader from any side, front or back.



➡️ Remove the Gearbox assembly to have access to the PCB, disconnecting the cables is recommended.



➡️ From the sides or from top/down edges pull gently the blue PCB that should came easily and leave the TTGO exposed.

Assembly :

This part of the guide will help you to assemble/reassemble the Nano Speedloader.

➡️ Plug in the Custom circuit board, it can be easily installed thanks to the header pins allowing easy install action and removal to the TTGO for programming.

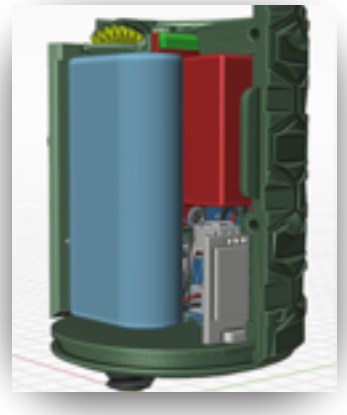


‼️ Please mind that Only 11 lower pins are used, make sure the two first pins are **not Plugged**. Future versions of the board will have 12 pins to reduce risk of mistake.

➡️ Once the Board is in place, **install the servo-gearbox assembly** inside the front housing using the **dedicated rails** and notches, assembly should slide into place without much friction.

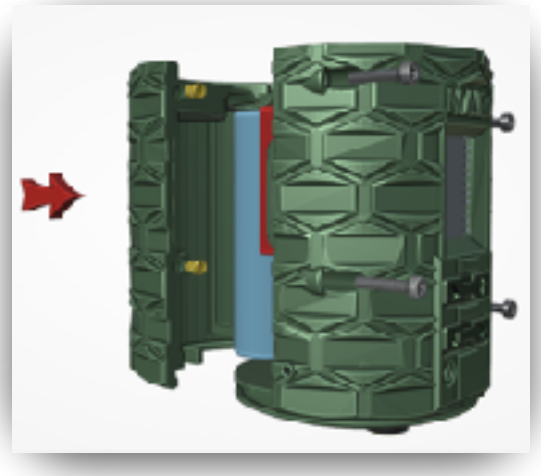
➡️ Plug the connexions, Each **connection is labelled** to avoid confusion.





➡️ Install the **BMS module**, the battery. Note that lover cover has a notch to maintain the USBC charger in place.

➡️ Slide **bottom cover** in place starting from the **trigger side** first then align the charger and let the **USB C** port get into position. Make sure the board is well positioned to let the Speedloader close properly.



➡️ Add the back shell to the assembly, it should slide in place easily. Add the four (4x) M3*16mm screws. If you feels resistance make sure there is no cable on the way..

‼️ **Do not over-tighten the screws**, as it might snap the shell in two. If the loader does not close properly, the might be cables on the way.

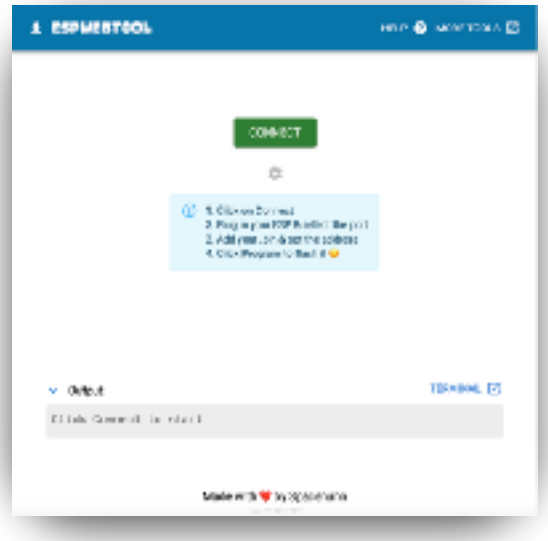
➡️ Last step is to **install the Extension** on the

Speed loader, it can be fixed quickly thanks to the **L shape lock** mechanism.



ℹ️ Use the **lock pictogram** to confirm that the mechanism is secured in place. Lock orientation is the same (clockwise) for the cover and the extension.

3 Programming 5 Min



The Board used on this project is new offering Over The Air (OTA) updates, allowing to update the firmware and via WIFI, from any Device (Mobile ready)

However, if you are running on software version V5x, the **first upload** need to be performed via **USB Cable**, made easy using the **ESP Web Tools**.

ℹ️ Firmware version is displayed at start up and within the settings submenu « About » on older versions, nothing is displayed,

⚠️ Only **Chrome** or **Edge** Browsers are supported.

➡️ Go to the following url :

https://esptool.spacehuhn.com/

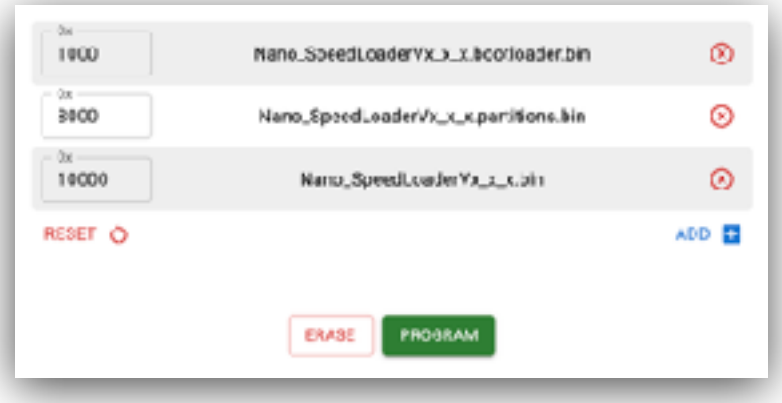
➡️ Connect your board to USB and click on **connect.** ⚠️ Make sure Arduino IDE is closed

➡️ Make sure each **address is properly entered** with corresponding file

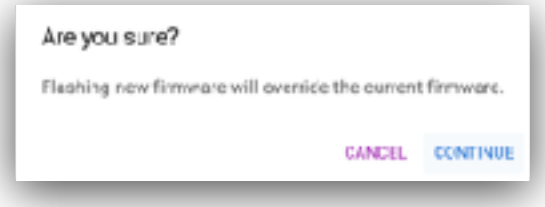
1000 : Nano_SpeedLoaderVx_x_x.bootloader.bin

8000: Nano_SpeedLoaderVx_x_x.partitions.bin

10000: Nano_SpeedLoaderVx_x_x.bin (main firmware)



➡️ Press **PROGRAM**, and **confirm** overriding the firmware





ℹ️ Loading process takes up to **2/3 minutes** for the very first upload. Once the upload complete, simply unplug the board, and you are **ready** 🎉!



WIFI Updates (OTA) :

For future releases, Updating the board will be very easy and via WIFI, on the NANO,
➡️ go to : **Settings** > **About** > **Long press on OTA**. (Follow instructions on the screen same as below)
ℹ️ The board will restart in **update mode**.

➡️ Using any device, connect to the **wifi network** named « NANO - UPDATE » and enter the password : « protoforge123 »

➡️ Go to URL : « **192.168.4.1** »

➡️ Select the firmware file and hit Upload, the board will take care for the rest then restart it self.

⚠️ Make sure to use the right firmware file (no boot loader neither partitions needed here) file to use : **`Nano_SpeedLoaderVx_x_x.bin (main firmware)`**

Contact and support :

protoforge.fr
protoforgeconcept@gmail.com